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Capitalistkind revisited: NewSpace, Elon Musk's SpaceX, and the politics of oligarchic infrastructure

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Abstract: This chapter examines the political economy of "NewSpace" through the case of Elon Musk's SpaceX. Incubated by public spending and government contracts, SpaceX has become a near-monopoly on which the U.S. state is dependent. Dominant narratives either overemphasize private entrepreneurialism, overlooking the state's foundational role, or fail to theorize the infrastructural mechanism by which private wealth is converted into political power. The concept of *oligarchic infrastructure* shows how structural dependence, personal control, and an inter-field conversion circuit give a private owner leverage over state and society. Revisiting the concept of capitalistkind and drawing on work on "oligarchic sovereignty," the chapter traces the discursive, material, and legal dimensions of shifts in the space economy, including the erosion of the Outer Space Treaty's commons framework, a process I term *cosmic enclosure*. What began as a state-incubated NewSpace has been eclipsed by "Big Space," a formation increasingly dominated by one man and one company.

Keywords: NewSpace; Oligarchic Infrastructure; Cosmic Enclosure; Capitalistkind; Infrastructural Power; Elon Musk; SpaceX

Introduction

In May 2026, SpaceX filed for a listing on the Nasdaq stock exchange, reportedly aiming for a valuation of $1.75 trillion (Wang, 2026). In an IPO prospectus filed with the U.S. Securities and Exchange Commission (SEC), SpaceX, whose founder and controlling shareholder is Elon Musk, who became the world's first trillionaire following the IPO and has since moved in and out of that status (Durot, 2026; Folk, 2026), claimed to have identified a market worth $28.5 trillion, "the largest actionable total addressable market ('TAM') in human history" (SpaceX, 2026). Only $370 billion, or around 1%, of this total, however, was attributed to "Space" proper, with a further $1.6 trillion to Starlink's satellite connectivity; the remaining $26.5 trillion, or more than 90% of its purported market, was attributed to artificial intelligence (AI). Its AI infrastructure ambitions included "orbital AI compute" (SpaceX, 2026), the technical and economic viability of which remains unproven (U.S. Government Accountability Office [GAO], 2026; Turyshev, 2026). The filing exemplified Musk's longstanding ambition to create a wide-ranging conglomerate at the frontier of technological development. Yet the market projections, and the weighting of AI in what was nominally a space company, also seemed to lend support to what Slobodian and Tarnoff (2026, p. 23) call Musk's "financial fabulism," an exercise in narrative construction: a firm whose consolidated accounts, recast to include xAI and X, showed a loss of nearly $5 billion the year prior (Mac & Hirsch, 2026) was now reportedly seeking a valuation of nearly a hundred times its annual revenue.

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The filing also showcased a company, and by extension its controlling shareholder, that held an increasingly dominant, if not near-monopolistic, position in the U.S. space-launch market. Through SpaceX's ascent, we can observe how one especially powerful, wealthy individual, Elon Musk, has come to accrue significant (privatized) infrastructural power (see, e.g., Atal et al., 2026; Bowen, 2026)—a potential veto power over a key sociotechnical domain—to the extent that the bounds between state authority and private corporate interests have increasingly blurred. While the historical sociologist Michael Mann (1984, 2008) coined the notion of "infrastructural power" as it pertained to the modern state, with the case of SpaceX we see a single corporate entity constructing and capturing essential (orbital) infrastructure on which the state is reliant—a corporate wedge into the state that, combined with Musk's other holdings in communications, AI, and industrial production, affords him significant political power, congealing into what Taşkale (2025) calls a "reactionary futurism." Building on the nascent concept of "oligarchic sovereignty" (Atal et al., 2026), this chapter develops this idea further with the notion of *oligarchic infrastructure*, denoting the material implementation of an intertwining of state necessity and corporate ownership, in which one hyper-wealthy individual has attained significant power through control over core infrastructure.

Oligarchic infrastructure, then, denotes a formation in which extremely wealthy private owners turn control over essential infrastructure into socio-political power. It is related to Riley and Brenner's (2022, p. 6) notion of political capitalism, in which "raw political power, rather than productive investment, is the key determinant of the rate of return," but it operates more at the level of particular actors than Riley and Brenner's economy-wide regime of accumulation, and it captures the obverse movement, describing instead the *state-like power that infrastructural control yields*. Certainly, the phenomenon is not new in and of itself—one finds analogues in private control over water, roads, food distribution, and more across both the Global South and the West, as well as historical parallels such as the formidable state-like authority of corporations like the East India Company (cf. Stern, 2011)—but its contemporary form is distinctively technological and digital. As proponents of the technofeudalism thesis have argued (Varoufakis, 2024; Wark, 2021), and as Bratton's (2015) notion of "The Stack" suggests, command over core digital infrastructure confers extraordinary leverage in a thoroughly technologized world. Musk's ever-widening portfolio illustrates this point with unusual clarity. SpaceX's launch dominance, sketched above, is only one infrastructural foothold. Alongside Starlink's satellite connectivity, the company has, since a February 2026 merger with xAI, also included the social media platform X—the $44 billion Musk acquisition, absorbed into xAI in 2025 (Bensinger, 2025), that constitutes a communications infrastructure doubling as a political influence apparatus (Schiffer, 2024)—and Grok's AI models (Grush et al., 2026), while Tesla extends Musk's reach further still into industrial production. The result is less a space firm than an increasingly multisectoral actor, controlled by an owner seemingly intent on constructing an "everything" portfolio—Musk has declared his desire for X to be an "everything app" (Milmo & Hawkins, 2023)—straddling manufacturing, social media, AI, space, and more.

Some years ago, a colleague and I coined the term *capitalistkind* to describe the commercialized turn in spaceflight that was gathering pace by the late 2010s (Shammas & Holen, 2019). Where the astronaut Neil Armstrong had proclaimed "one giant leap for mankind" upon setting foot on the lunar surface in 1969, it seemed increasingly as though the new entrepreneurs of the (ostensibly) privatized space economy were making "giant leaps" of their own—only this time for their enterprises, and for their private fortunes. Billionaires Elon Musk, Jeff Bezos, and Richard Branson had entered a new space race, framing their ambitions in conspicuously humanist-universalist terms: from Musk's (2017, p. 46) musings about

turning humanity into a "multi-planetary species," a sort of collective insurance policy against "some eventual extinction event," to Branson and Bezos's promises of affordable space tourism. But the universalist packaging was more distraction than reality: with the multiplication of space launches, the atmospheric emissions from launches and re-entry space debris might instead, ironically, compound the very planetary risks Musk claimed to be guarding against (Ferreira et al., 2024; Ryan et al., 2022). By the late 2010s, unlike in 1969, it was not "mankind" that was taking the next step into the cosmos, but a particular circle of wealthy capitalists, who nevertheless mobilized humanist narratives to bolster their activities—denoted in shorthand with the term *capitalistkind*.

Since that article was published in 2019, what we described has intensified considerably. What was optimistically termed "NewSpace"—the idea of unencumbered entrepreneurship as a source of renewal in the space sector, set against the supposedly outmoded "OldSpace" dominated by state agencies like NASA—has in practice led to the emergence of a single private actor wielding an unprecedented degree of dominance in outer space. In 2025, SpaceX conducted 165 orbital launches, accounting for approximately 85% of all U.S. launches that year (Wall, 2025), more than half of all global launches and more than 80% of all payload mass carried into space that year (BryceTech, 2026). The company's Starlink satellite constellation encompassed some 9,600 active satellites—or "approximately 75% of all active maneuverable satellites in orbit" as of March 31, 2026 (SpaceX, 2026, Prospectus Summary)—with an application for up to a million more satellites as orbital data centers (Davies, 2026). The scale of its megaconstellation has led some scholars to ask whether SpaceX in effect "occupies" low Earth orbit (LEO) (Bennett & Cudney, 2026). SpaceX is not so much a representative case as a *leading* case, the most advanced member of a formation toward which other private space actors—most notably Bezos's Blue Origin—aspire, and one in which the mechanism by which infrastructural control is converted into political power is therefore easiest to see.

This chapter argues that the dominant narratives surrounding the commercialization of outer space—whether celebratory accounts of "NewSpace" entrepreneurialism or critical analyses focused narrowly on "billionaires in space"—are both incomplete. What we are witnessing is not so much "privatization" in any straightforward sense as the *personalization* of a long-standing hybrid state–corporate space regime: this political-economic formation, in which commercial and state logics are structurally interwoven, is increasingly dominated by a single owner-controlled firm. NewSpace, in other words, has given way to what we might call Big Space. Meanwhile, the semi-idealistic legal architecture that sprang up around the notion of outer space as a global commons is being reconfigured, increasingly by multilateral consensus, along extractivist lines better aligned with dominant actors' geopolitical interests and the interests of private capital. I label the formation at the center of this shift *oligarchic infrastructure*. *Capitalistkind* is at once the assemblage of key private space enterprise actors and the (falsely) universalizing discourse that legitimates this assemblage, while *cosmic enclosure* is the legal architecture that entrenches oligarchic infrastructure, extending its capacity for resource extraction in space.

This chapter proceeds through five sections. Section 1 revisits the concept of capitalistkind (Shammas & Holen, 2019), reconstructing the commercialization narrative from the Obama-era SPACE Act through the first Trump administration, and argues that the opposition between "OldSpace" and "NewSpace" was always more ideological than empirical. Section 2 draws on Mazzucato's (2014) entrepreneurial state thesis to show how SpaceX is a product of intensive state investment, and how the company's dominance has since made the state depend on it. Section 3 develops the concept of *oligarchic infrastructure* and specifies its three dimensions:

structural dependence, personal control, and an inter-field conversion circuit of power. Section 4 turns to "cosmic enclosure," showing how the ever-expanding Artemis Accords and, increasingly, draft multilateral principles reinterpret the Outer Space Treaty to allow resource extraction while leaving benefit-sharing voluntary, reworking the implications of a "global commons" framework from within it, thereby reinforcing oligarchic infrastructure. Finally, Section 5 sketches a research agenda for the critical political economy of outer space and considers the limits of the argument.

1. Capitalistkind revisited

In the late 2010s, private enterprise had become an increasingly prominent factor in outer-space discourse. The rise of NewSpace was aided by Obama-era legislation: the U.S. Commercial Space Launch Competitiveness Act (2015), or SPACE Act, aimed to "facilitate a pro-growth environment for the developing commercial space industry by encouraging private sector investment" and granted private U.S. entities the right to "any asteroid resource or space resource obtained, including to possess, own, transport, use, and sell" these (§ 402). While the Act's Title IV section on "Space Resource Commercial Exploration and Utilization" was framed as bounded by relevant international obligations, this was also a significant marketization of space activities and a departure from the Outer Space Treaty's egalitarian "commons"-centric view of space. A growing chorus of voices celebrating "NewSpace" grew up in the wake of this legislation. This largely pro-NewSpace discourse hailed the privatization of outer space activities as a fount of innovation, reduced cost, and accessibility. People like Harvard Business School's Matthew Weinzierl welcomed NewSpace: "The idea of leveraging the private sector and its efficiencies to help achieve public sector goals is a powerful idea for the students to engage with," Weinzierl said in presenting a 2016 HBS case study on Blue Origin (Camerota, 2016). "It helps that we're at a remarkable point in history where we have people like Jeff Bezos and Elon Musk with the vision and the capital to even make such a discussion possible." Weinzierl would later publish a book, *Space to Grow: Unlocking the Final Economic Frontier*, that extolled the "revolution" of NewSpace, now framed as the *addition* of "market forces" to government's continuing role; "not a removal of the public sector's vital role in space but an integration of the private sector" (Weinzierl & Rosseau, 2025, p. 1).

The moderate Obama-era enthusiasm for space commercialization accelerated significantly under the first Trump administration. Trump's 2018 memorandum, Space Policy Directive-2, was a classic attempt to liberalize market regulations, ordering the government to "examine all existing…requirements, standards, and policies associated with commercial space flight launch and re entry [sic] operations" with the aim of "minimiz[ing] those requirements, except those necessary to protect public safety and national security" (White House, 2018). This deregulatory impetus was accompanied by a significant discursive shift that increasingly repudiated the commons-oriented framework that, nominally at least, had governed outer space since the 1960s. The year prior, the American Space Commerce Free Enterprise Act, a U.S. House of Representatives bill, had unilaterally declared that "outer space shall not be considered a global commons" (2017, § 80308). While the bill ultimately failed, an academic analysis at the time rightly diagnosed it as constituting a political-legal attempt to dismantle the notion of space as a global commons (Hao & Tronchetti, 2019).

The executive director of the National Space Council, Scott Pace, publicly reinforced this position, proclaiming that "outer space is not a 'global commons,' not the 'common heritage of mankind,' not 'res communis,' nor is it a public good" (Pace, 2017, p. 4). What had been axiomatic to international space governance since the late 1960s was being rhetorically

dismantled by senior Trump officials. Speaking before the National Space Council in February 2018, Secretary of Commerce Wilbur Ross noted approvingly that space was "already a $330 billion industry" poised to become "a multitrillion-dollar one in coming decades," and declared that it was "time to unshackle business activity in space" (U.S. Department of Commerce, 2018). Ross even toyed with the idea of turning the Moon into a "gas station for outer space" (Bryan, 2018), distilling the entrepreneurial imaginary that was rapidly unfolding under Trump 1.0.

Surrounding and reinforcing this policy discourse was a broader speculative frenzy about the fortunes to be made beyond Earth's atmosphere. Industry commentators and financial analysts projected that the space economy would produce the world's first trillionaires (Honan, 2011); one early study of asteroid mining (Lewis, 1996) gained prominence when Silicon Valley entrepreneur Peter Diamandis cited Lewis's work in claiming, at a 2006 space conference, that "there are twenty-trillion-dollar checks up there, waiting to be cashed!" (Honan, 2011). This speculative imaginary, in which outer space figured as an essentially limitless reservoir of extractable value, served an important legitimating function, reframing the commercialization of the "commons" of outer space not as a process of enclosure but as entrepreneurial opportunity, a rational exploitation of a pioneering frontier. By December 2020, with Trump's first period drawing to a close, a newly formed U.S. National Space Policy (White House, 2020b) proclaimed a desire to "create an environment that energizes our industry to create innovative commercial approaches" (p. 1) and held aloft as one of its core principles the importance of a "robust, innovative, and competitive commercial space sector" (p. 3).

Central to NewSpace discourse was a clear-cut binary: OldSpace—the supposedly plodding, state-centered manner of relating to outer space—versus NewSpace, a sleeker, market-driven paradigm of entrepreneurial agility. The notion of a private-led "Second Space Age" (Berger, 2024) leading into the "greatest business opportunity of our lifetime," as Chad Anderson's (2023) *The Space Economy* promises, has proven highly influential as a narrative framing, both in popular and (semi-)academic discourse (for a critical analysis of these "sociotechnical imaginaries," see Tutton, 2021). The discursive opposition simultaneously mirrored the broader post-Cold War narrative of a transition from Keynesian-Fordist statism to a neoliberal, market-oriented order (Harvey, 2005). But the binary was always more ideological than empirical. Private enterprise had played a significant role in U.S. space activities from the outset—the Apollo Lunar Module, for instance, was manufactured by the aerospace contractor Grumman, albeit with public financing and under state direction. Conversely, NewSpace was never as "new" as its proponents claimed: private companies like SpaceX derived a significant proportion of their early revenue from government funding and public contracts; one report estimates that SpaceX "operated on total funding of approximately $1 billion in its first ten years of operation, about half of which came from progress payments on government contracts" (Space Angels, 2019, p. 2).

OldSpace was never old; NewSpace was never new. The private sector was always involved; the state never departed. Just as scholars of neoliberalism have demonstrated that neoliberalism entails not the minimization of the state and the maximization of markets but their mutual imbrication, with the former pressed into service for the latter (Wacquant, 2012), scholars of outer space ought to recognize the hybridization of state and market actors in this domain as a given. A company like SpaceX enjoys the benefits of a favorable regulatory environment and public contracts. But with the consolidation of SpaceX's near-monopoly in U.S. launch activities, what has changed is who holds leverage within this relationship: the balance of power has shifted from the state to a single firm and a single owner.

2. From state investment to private power

Mazzucato's (2014) concept of the "entrepreneurial state" holds that private-sector accomplishments are often the result of public investment and state-led risk-taking and market creation. From pharmaceuticals to the Internet, the state funds the basic research and builds the foundational infrastructure that no private actor could on their own. The space industry is no exception. Examining NASA's relationship with private enterprise, Mazzucato and Robinson (2018) highlighted how the agency was moving from a central, directing role toward a more facilitating role; they raised concerns about NASA's capacity to "direct change" and set the "directionality" of space enterprise (Mazzucato & Robinson, 2018, p. 175). Their concerns were more than justified; what they did not explicitly anticipate was how dependent NASA would become on a single private firm it helped nurture into being. SpaceX, incubated by NASA and Pentagon contracts, grew to dominate the U.S. orbital system so completely that by 2025 NASA and the U.S. Department of Defense had grown heavily reliant on it. The entrepreneurial state had built a private quasi-monopoly, and that actor could now exert considerable leverage over its erstwhile patrons as well as the wider state and society surrounding it.

Despite the privatization thesis advanced by proponents of NewSpace and by the first Trump administration alike, the state did not, of course, withdraw from outer space activities. SpaceX was incubated and nourished by the state at every step of the way. A $1.6 billion Commercial Resupply Services contract awarded by NASA in late 2008 rescued the company at a crucial juncture. As Vance (2015, p. 210) recounts, contemporary press reports suggested that SpaceX, "the onetime front-runner for the large NASA contract, had suddenly lost favor with the space agency," and that this contract therefore represented a dramatic reversal of fortunes. A 2025 Washington Post investigation found that Musk's companies had benefited from at least $38 billion in government contracts, loans, subsidies, and regulatory credits over more than two decades, with SpaceX accounting for the largest share, approximately $22.6 billion (Butler et al., 2025); importantly, "the total amount is probably larger," as the Post's analysis only includes "publicly available contracts, omitting classified defense and intelligence work for the federal government"—a particularly salient omission given SpaceX's involvement in national-security work. Nor has this support ended: SpaceX continues to derive a substantial portion of its revenue from NASA, Pentagon, and national security-related contracts, including $6.45 billion in Space Force contracts in May 2026 (O'Kane, 2026). The Randian image of heroic entrepreneurship that Musk has long cultivated, along with a techno-libertarian reading of largely progressive science-fiction novels (Boucher & McAvan, 2026), fits uneasily with these material realities.

The U.S. government was SpaceX's single largest customer in 2025, yet it accounted for just over 20% of the company's revenue (SpaceX, 2026), suggesting that dependency in the state–corporate relationship now runs deeper in the opposite direction. With some four-fifths of its revenue coming from other sources, notably Starlink's commercial subscribers, SpaceX is much less reliant on government contracts than it was in earlier years: it now needs the U.S. state far less than the state needs it. Neither United Launch Alliance (ULA), a Boeing–Lockheed Martin joint venture, nor Jeff Bezos's Blue Origin approaches SpaceX's launch cadence, so the U.S. government could not easily replace SpaceX without a significant degradation of launch capabilities. Growing great-power rivalry only increases this asymmetry: as access to outer space becomes more critical to national security and power projection, the risks of relying on a single private provider grow, but so does the price of a capability gap while alternatives are built. Because these can take years to develop—and Blue Origin's New

Glenn rocket explosion during a ground test in May 2026 was a dramatic reminder of the technical difficulties involved (Rosner-Uddin, 2026)—the state may come to tolerate ever-increasing short-term dependence even as concerns mount, pursuing diversification only over a longer timespan. The Starlink episodes in Ukraine, discussed next, illustrate both sides of this dynamic: the political risks of oligarchic infrastructural control are evident, yet state reliance continues.

Starlink's role in the Russia–Ukraine war probably offers the clearest encapsulation of what oligarchic infrastructural leverage looks like in practice: during a Ukrainian drone attack against the Russian Black Sea Fleet in September 2022, operators lost Starlink connectivity at a critical moment. The precise sequence of events remains contested—Musk claimed the area around Crimea had been geofenced prior to the attack, while his biographer Walter Isaacson (2023) initially reported, then retracted, the claim that Musk had actively deactivated coverage (Darcy, 2023; see also Abels, 2024). Four years later, Starlink's privatized geopolitical role was again the subject of public outcry. In October 2026, during a dispute between Washington and Kyiv over Ukrainian strikes on Russian oil infrastructure, Ukraine's access to both Starlink and the military-grade Starshield satellite service was restricted, interfering, on Kyiv's account, with its defensive operations. Musk first denied throttling satellite connectivity to Ukraine's defense forces, then claimed SpaceX had closed a "loophole" after Ukraine's military allegedly "violated Starlink terms of service," while attributing responsibility to the U.S. government for any changes to Starshield service; satellite connectivity was eventually restored (Blann & Morton, 2026; Phillips, 2026). It is unclear whether Musk was acting independently or in conjunction with the Trump administration: Trump said he had told Musk to "put it back" while denying he had ordered the restriction; Vice President JD Vance "hinted at White House coordination," while Secretary of State Marco Rubio and Defense Secretary Pete Hegseth "denied any administration involvement" (Blann & Morton, 2026). These episodes show a private firm intervening in the operational parameters of a sovereign state's defensive efforts, or what Muggah and Glenny (2026) describe as a "privatized geopolitics" in which "a private company is now a gatekeeper in orbit."

Alongside infrastructural leverage, Musk has also exercised political and communicative power directly. In 2022, he acquired Twitter (now X) for $44 billion, gaining control over a consequential social media network, still the world's leading microblogging platform. A Bloomberg investigation ahead of the 2024 U.S. presidential election found that Musk had become the single most important disseminator of anti-immigrant conspiracy theories on X (Alba et al., 2024). He spent some $288 million in the 2024 presidential election in support of Trump and other Republican candidates (Thadani et al., 2025) and subsequently assumed a leading role in the newly created Department of Government Efficiency (DOGE), charged with implementing sweeping cuts to federal agencies—even as, in April 2025, his company SpaceX was awarded a $5.9 billion Space Force contract (Lemon & Mesa, 2025).

SpaceX's May 2026 filing followed its February 2026 absorption of Musk's AI venture xAI—which had itself absorbed the social media platform X and the chatbot Grok—in a deal valuing xAI at $250 billion (Grush et al., 2026), creating a single corporate entity that unified orbital launch, satellite communications, social media, and AI. Bloomberg noted that analysts were being asked to value the company not on present revenue but on "fear of missing out" and speculative "optionalities" (Lipschultz, 2026); that is, future capabilities not yet realized: the frenzy of "fictitious capital" (Durand, 2017) attempting to make itself real through the symbolic operations of extreme narrative construction. Not all investors were persuaded by this offering: AkademikerPension, the Danish pension fund managing $25 billion in assets, publicly refused

to participate, describing SpaceX as "grossly overvalued" with what its chief investment officer called a "catastrophic governance structure," noting that Musk would retain more than 80% of voting rights while serving simultaneously as CEO, CTO, and board chairman (Schwartzkopff, 2026).

3. Oligarchic infrastructure

Mann distinguishes two forms of state power: *despotic*—the power to make decisions "without consultation with civil society groups" (Mann, 2008, p. 355)—and *infrastructural*, the capacity of the state to "actually penetrate civil society and implement its actions" (p. 355) through systems of taxation, roads, railways, and other essential services. The former predominated in early-modern societies, the latter in modern ones: "[M]odern capitalist democracies are despotically weak—leaders rule only with the consent of representatives either of the people or of the dominant classes—but they have more infrastructural power" (2008, p. 356). In Mann's (1984) original typology, the authoritarian state, on the other hand, is defined by a combination of high infrastructural and high despotic power (p. 191), capable both of penetrating society *and* implementing decisions "without routine, institutionalised negotiation" (p. 188). Oligarchic infrastructure reproduces this combination, only now in private hands: the "despotism of capital" described by Marx (1867/1976, p. 526), fused with a grip on infrastructure that the state and society require. Mann's account of state autonomy is illuminating here: those who control essential centralized services can exploit the "social utility" of those services, so that "from collective power there derive[s] a measure of distributive power" (2008, p. 355), that is, a *shared* capacity or service turning, for whoever controls it, into a power *over others*. Partaking of the logic of both of Mann's forms of (state) power, an individual corporation-and-owner—an assemblage of personalistic corporate power—is thus able to convert infrastructural power into distributive power, implementing a particular vision, a politics, and a set of practices, bounded only uneasily by regulatory frameworks and political structures, in an essentially nondemocratic manner.

Certainly, there are numerous corporations that wield significant infrastructural power, a power that can be described as corporate-monopolistic. In the digital domain alone, Alphabet/Google remains dominant within search; Amazon's AWS has captured a large share of the server market, serving as a digital backbone to significant portions of the Internet; Meta commands control over the platforms that serve up content to billions of social media users. What distinguishes the case of Musk, and by extension SpaceX, is the combination of state dependence, personal control, and political influence that this infrastructure has yielded. By 2025, SpaceX was the primary commercial provider of crewed spaceflight to the International Space Station, the dominant launch provider for national security payloads, and the operator of a satellite constellation on which the U.S. military increasingly relies for communications and sensing in ongoing conflicts. ULA, Blue Origin, and other actors could, in principle and in time, wean the state off this dependence. Yet at present, the U.S. capacity to project power via outer space increasingly relies upon the hardware and knowhow of a single private corporate actor, which thereby occupies what Farrell and Newman (2019) call a chokepoint, a position of leverage from which access to a "network hub" (p. 56), and the core state or societal functions that flow through it, can be granted or withheld—here, however, unlike in Farrell and Newman's analysis, the chokepoint is being held *over* the state, not by it. Arguably, it is this dependency combined with Musk's ideological realignment, considerable campaign donations, and communicative power via the $44 billion acquisition of Twitter that facilitated the most striking feature of this case study: his installation within the executive branch itself, through DOGE, while his companies remained subject to oversight by the very agencies he was tasked with restructuring.

Oligarchic infrastructure, specified more precisely, is a formation characterized by three dimensions: (1) structural dependence: concentrated control over an infrastructural system or asset on which the state relies for some core functionality and for which substitutes are slow, costly, and difficult to develop, such that disciplining the private actor in question would risk degrading a state's core capacities; (2) personal control: ownership and decision-making authority concentrated in a single owner, frequently insulated from checks by shareholders, market actors, and other stakeholders; and (3) a (Bourdieusian) inter-field conversion circuit linking economic, infrastructural, communicative, and political forms of capital, through which the figure of the owner-oligarch can convert different forms of power into one another.[2] In extreme form, this circuit can even grasp after what Bourdieu (2012/2014) calls the state's "meta-capital," its unique "power above powers" (p. 197), which is to say its capacity to determine the value of all other capitals (see also Shammas, 2018). Musk's installation in DOGE, discussed above, marks one such extreme point, if only temporarily and with partial access to such meta-capital. When these dimensions converge, they tend to weaken governance mechanisms (e.g., regulatory agencies, environmental protections, labor rights, and multilateral frameworks) that might impose external discipline and accountability on the actor in question, which only further entrenches oligarchic infrastructure's position of leverage.

There is nothing new about capitalism's oligarchic tendencies nor the tendency toward concentration of economic ownership—and by implication, political influence—in the hands of a diminishing subset of capital owners. In the nineteenth century, Marx described the law-like regularity of the "centralization of capitals," the dynamic by which "one capitalist always strikes down many others" (1867/1976, p. 929)—more figurative than literal in late modernity, to be sure. In the twentieth century, Baran and Sweezy (1966) formalized this tendency toward "monopoly capitalism." The fact of heightened concentration of ownership into "giant corporations" has, on their account, definite political implications: as they write of the United States, "in constitutional theory, the people exercise sovereign power; in actual practice, a relatively small moneyed oligarchy rules supreme" (p. 157), a diagnosis echoed in Senator Bernie Sanders's "Fighting Oligarchy" campaign in our own day (see Sanders, 2025). Winters (2011) conceptualizes *oligarchs* as actors who "command and control massive concentrations of material resources that can be deployed to defend or enhance their personal wealth and exclusive social position" (p. 6). *Oligarchy*, in turn, "refers to the politics of wealth defense by materially endowed actors" (p. 7). In the U.S., which Winters labels a "civil oligarchy" in his typology, the extremely wealthy submit to the state—they are "fully disarmed" and "do not rule directly" (p. 36)—in exchange for "wealth defense" (pp. 20–26), the state's protection of property and income. The concept of oligarchic infrastructure reorients and broadens this configuration; rather than depending on the state to defend their wealth, oligarchs become actors on whom the state depends and who can pursue agendas—technological, political, social, and cultural—in excess of mere wealth defense, even as this undoubtedly remains important to them. It thereby adds a material mechanism to this theoretical lineage by positing that not wealth alone, but control over societally critical infrastructure that such wealth either secures or accompanies, is essential to understanding how power operates.

[2] In Pierre Bourdieu's field theory, fields are semi-autonomous domains where agents compete to produce bounded, field-specific outputs and to dominate the respective field to which they belong (Bourdieu & Wacquant, 1992). These semi-enclosed spaces, which are the result of the functional differentiation of social life, respect specific logics and recognize different capitals, which are often integral to the field in question. What characterizes the new tech oligarchy is its capacity to operate across a range of different fields, including the span of what Bourdieu termed the "field of power," integrating multiple forms of power.

Recent scholarship has theorized this broader phenomenon as "oligarchic sovereignty," the claim by tech oligarchs to quasi-sovereign authority (Atal et al., 2026). Lockwood, in Atal et al. (2026), draws on Mann's concept of infrastructural power to analyze the fusing of corporate, personal, and state authority. Similarly, Hahamovitch et al. (2026) have traced the genealogy of the tech oligarchy within the space sector as a product of long-standing capitalist technoscience. These accounts form part of a wider literature on the political effects of privately owned infrastructure. Work on platform capitalism argues that firms that control digital infrastructure wield significant leverage over state and society (Rahman & Thelen, 2019); indeed, Lehdonvirta (2022) shows how platforms increasingly have come to assume state-like authority. Scholarship on assetization and rentier capitalism, meanwhile, shows how control over assets has enabled the capture of significant revenue streams (Birch & Muniesa, 2020; Christophers, 2020). Drawing on these literatures, we can think of orbital launch capabilities as a kind of "platform," an *interface* between Earth and space, or a "chokepoint" from which significant concessions can be extracted.

The concept of oligarchic infrastructure, as developed in this chapter, is a systematizing complement to these literatures. It zeroes in on the material-infrastructural mechanism through which tech oligarchs' claims on a particular political status are actualized. It specifies the conditions under which infrastructural control places political leverage in the hands of a single firm and owner. The three dimensions outlined above—structural dependence, personal control, and an inter-field conversion circuit—provide a framework for examining this concentration of leverage and its potential to weaken governance mechanisms. Slobodian and Tarnoff (2026) have termed this broader political formation "Muskism," in which techno-sovereignty is sold to states and individuals alike, binding them into infrastructural dependency on a single actor: "Both the Pentagon and NASA depend on SpaceX; Starlink has become indispensable on the battlefield […]; X and Grok are being woven into the state. Trying to unplug from Musk, you realize he owns the socket" (p. viii).

4. Enclosing the cosmic "global commons"

The first section of this chapter traced the discursive reconfiguration of the commons-oriented view of outer space in the world's dominant spacefaring nation; this section examines the legal architecture that is being established to support this move. The legal point of departure is the 1967 Outer Space Treaty (OST), which holds that the exploration and use of outer space shall be "carried out for the benefit and in the interests of all countries" and be "the province of all mankind," and outer space itself is not to be "subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means" (Outer Space Treaty, 1967, Art. I–II). While this multilateral framework never explicitly used the term "global commons," the interpretation of the 1967 OST as a commons regime was the predominant one (Buck, 1998; Freeman, 2025)—a commons in the sense of open access and non-appropriation, rather than the "common heritage of mankind" with "equitable sharing" of benefits that the 1979 Moon Agreement would later try, largely unsuccessfully, to establish (Vogler, 2000, p. 100). For decades, this framework was foundational to the global understanding of space, perhaps not so much because states were deeply committed to its egalitarian ideals—though there was a certain nonaligned idealism about its 1960s inception—but because much of space remained materially inaccessible to even dominant states and extractive capital. The commons principle was essentially cost-free to uphold so long as there was nothing, in practical terms, to enclose or extract. As the technological and commercial conditions for outer-space exploitation have shifted, so too have the ideological environment and legal architecture.

In April 2020, Trump signed an executive order directing federal agencies to promote "the public and private recovery and use of resources in outer space" and stated bluntly that "the United States does not view [outer space] as a global commons" (White House, 2020a). The order prepared the way for the first Trump administration's signature contribution to space governance, the Artemis Accords, a U.S.-led set of nonbinding principles—a "political commitment," as the Accords put it (NASA, 2020, § 1)—released in October 2020, just weeks before the presidential election that would bring Joe Biden to office; the Biden administration, notably, continued and expanded the initiative. Signed by 76 countries as of September 2026 (NASA, 2026a), the Artemis Accords are an exercise in juridical double-coded talk. They profess continuity with the OST, "desiring to implement" its provisions and affirming the "importance of compliance" with it (NASA, 2020, preamble). A critical provision in Section 10, however, affirms that the "extraction of space resources does not inherently constitute national appropriation under Article II of the Outer Space Treaty" (§ 10.2). This clause distinguishes between *national appropriation* of celestial bodies, which is prohibited by the OST, and *commercial extraction* of resources in space, which is treated as compatible with the OST—thereby legitimating private resource extraction while purportedly preserving the letter of the original space treaty. While this interpretation is not original to the Accords—the SPACE Act had already granted U.S. entities rights to extracted resources while disclaiming sovereignty over any celestial body (U.S. Commercial Space Launch Competitiveness Act, 2015, § 403)—the Accords in effect internationalized the legal sanctioning of space resource extraction as compatible with the UN treaty, pushing this interpretation to a growing coalition of signatories.

To be sure, the OST does not clearly prohibit the extraction of resources in outer space, while the Moon Agreement (1979), which envisioned a benefit-sharing mechanism for resource extraction on the Moon, has only 17 parties. And the Artemis Accords' interpretation of the OST may soon be more than a U.S. position or that of the Accords' growing list of signatories: draft principles of the UN Committee on the Peaceful Uses of Outer Space (COPUOS) for "space resource activities," still under negotiation ahead of an April 2027 session (Ünüvar, 2026), would, if adopted, legitimize the U.S.-led interpretation advanced in the Artemis Accords (COPUOS, 2026, Principle 2), facilitating private extraction in space by explicitly extending the freedom to carry out resource activities to "non-governmental entities" (Principle 2(A)). The September 2026 draft principles suggest a growing global consensus favoring enclosure in space, paradoxically, *within* a multilateral (putative) commons framework, in which no revenue-sharing is required: the draft only encourages states, "on a voluntary basis," to "consider sharing benefits" from space resource extraction with other states (Principle 9(C)). Moreover, the (non-exhaustive) examples are all non-monetary, including "scientific exchange" and "capacity-building," while revenue goes unmentioned. This *enclosure with a commons face* would, if adopted, strengthen the international acceptance of private resource extraction in space, while leaving the question of who benefits to the few states and firms capable of such activities.

What is unfolding, in short, is what Marx would have recognized as an "enclosure of the commons," the process by which shared resources are fenced off, appropriated, and turned into sites of capital accumulation (see, e.g., Harvey, 2003; for an updated appraisal of how this relates to the outer space economy specifically, see also Dy & Roberts, 2025). The historical enclosures of public land, or the "commons," in early-modern England dispossessed peasant communities while giving rise to the concentrated ownership of land and consequent wealth that fueled early capitalism (Marx, 1867/1976, Chapters 26–27). The cosmic enclosure now underway operates by analogous logic, albeit at cosmic scale: a legal architecture comprising

executive orders, domestic legislation, quasi-multilateral accords, and now draft multilateral principles is converting what the OST declared "the province of all mankind" into a potential domain of commercial extraction and private infrastructural control. This legal architecture in effect entrenches oligarchic infrastructure: NASA's planned return to the lunar surface depends on landers designed and built by Musk's SpaceX and Bezos's Blue Origin (NASA, 2026b).

5. Toward a critical political economy of outer space

The concept of oligarchic infrastructure developed above raises multiple possible questions for a research program in the critical political economy of outer space. First, under what conditions does a state's use of privately owned infrastructure turn into a relationship of dependence that can be leveraged politically by the owners of that infrastructure? State and societal reliance on such infrastructure suggests a potential privatized veto power, but its direct exercise must be scrutinized empirically on a case-by-case basis. The two Starlink episodes in Ukraine discussed above could be one starting point for such inquiry in the case of SpaceX, but a systematic assessment of the exact nature of state–corporate relations, including disputes, negotiations, concessions, regulatory outcomes, and cases in which the state nevertheless prevailed, remains to be undertaken. The concept opens a systematic inquiry into the *latent bargaining power* of powerful economic agents—to be distinguished from its *actual exercise*.

Second, to what extent can the state reduce its dependence on oligarchic infrastructure by stimulating competition, designing procurement processes, and retaining expertise, to limit corporate power? Can policies, contracts, and funding lines be designed in such a way as to ensure that public agencies remain in control, ensuring a proper hierarchy between the state as representative of social universality and private service providers? We should not be overly optimistic about these prospects: Bezos's New Glenn setbacks notwithstanding, among the most credible alternatives to SpaceX and Starlink are Bezos's Blue Origin and Amazon's Leo constellation (formerly Project Kuiper), the latter belonging to a company of which Bezos, a billionaire with oligarchic characteristics, remains executive chair and an important shareholder. Competitive diversification may dilute the power of any one oligarch but could also in aggregate multiply instances of oligarchic infrastructure rather than diminish them, replacing a single dominant owner like Musk with a circle of such tycoons.

Third, how solid is oligarchic infrastructure when it relies on speculative financial narratives to bolster itself? As noted above, SpaceX's IPO prospectus envisions a market worth $28.5 trillion; but this projection, and the valuation built on it, rests in large part not on present or even probable revenue streams but on speculative assessments about as-yet untested technology and uncertain future profits—a narrative construction or fictional imaginary straining to make itself real. What happens when this narrative runs up against the hard limits of engineering and economic constraints, in which projected markets do not materialize, or do so at a fraction of their projected scale? Researchers should ask who will be made to bear the costs of hard corrections to such narratives. Will the resulting financial distress weaken oligarchy or, paradoxically, deepen state involvement, raising costs to the public, as public agencies mobilize to prop up the owners of infrastructural capabilities that they have become dependent on? In short, is oligarchic infrastructure at once too overnarrativized to survive and too big to fail?

Fourth, what are the prospects for what we might call counter-enclosure? The Outer Space Treaty remains in force, and with it, the principle that the exploration and use of outer space should be "the province of all mankind." Researchers might ask under what conditions less powerful, non-spacefaring nations, which do not necessarily have unified interests, could build

coalitions to secure meaningful access to space and a share of its benefits, including profit-sharing stemming from commercial activities in what many still consider to be a "global commons." A critical political economy of outer space should also attend to the significant, and accelerating, negative environmental externalities of commercial activities, including climate-warming soot from launch emissions and the ozone-depleting particulates and metallic oxides released by obsolescent, re-entering satellites (Ferreira et al., 2024; Miraux, 2022; Ryan et al., 2022); these costs are borne globally, by all, yet the gains accrue only to a small number of states and corporations.

These questions also point to the limits of this chapter, which centers on the United States and a single (albeit leading) firm. The personalizing political power of oligarchic infrastructure may be a specifically U.S. formation, indicative of this society's unusually high tolerance for the private concentration of wealth and power. In countries or regions where launch capabilities and satellite operations are more firmly under state control or public direction, including China or the European Union, infrastructural control may be less likely to assume a privatized, oligarchic form. Oligarchic infrastructure requires a state willing to cede core capabilities to private owners. How far this is a case of U.S. exceptionalism—though of great global import, given U.S. centrality in space—remains an empirical question.

When Neil Armstrong spoke of taking "one giant leap for mankind," a universalistic aspiration, however ideologically freighted, was at least formally fused with a multilateral framework that declared the exploration and use of outer space "the province of all mankind." What this chapter reveals is not the disappearance of universalist rhetoric—SpaceX's S-1 claims its actions are ushering in the "next paradigm shift for humanity" and "contributing to the safeguards of humanity's future against existential risk" (SpaceX, 2026, Prospectus Summary); the Artemis Accords still position themselves as being in continuity with the Outer Space Treaty—but its progressive hollowing out. While the rhetoric remains in place, the legal, institutional, and material infrastructure around it is being restructured to serve a considerably narrower range of interests and set of actors. What began as a state-incubated NewSpace has given way to Big Space, a formation dominated by one company and the world's wealthiest person. It is not mankind but capitalistkind—and, increasingly, a single oligarchic actor within it—that sets the terms by which humanity's relationship to outer space is unfolding.

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